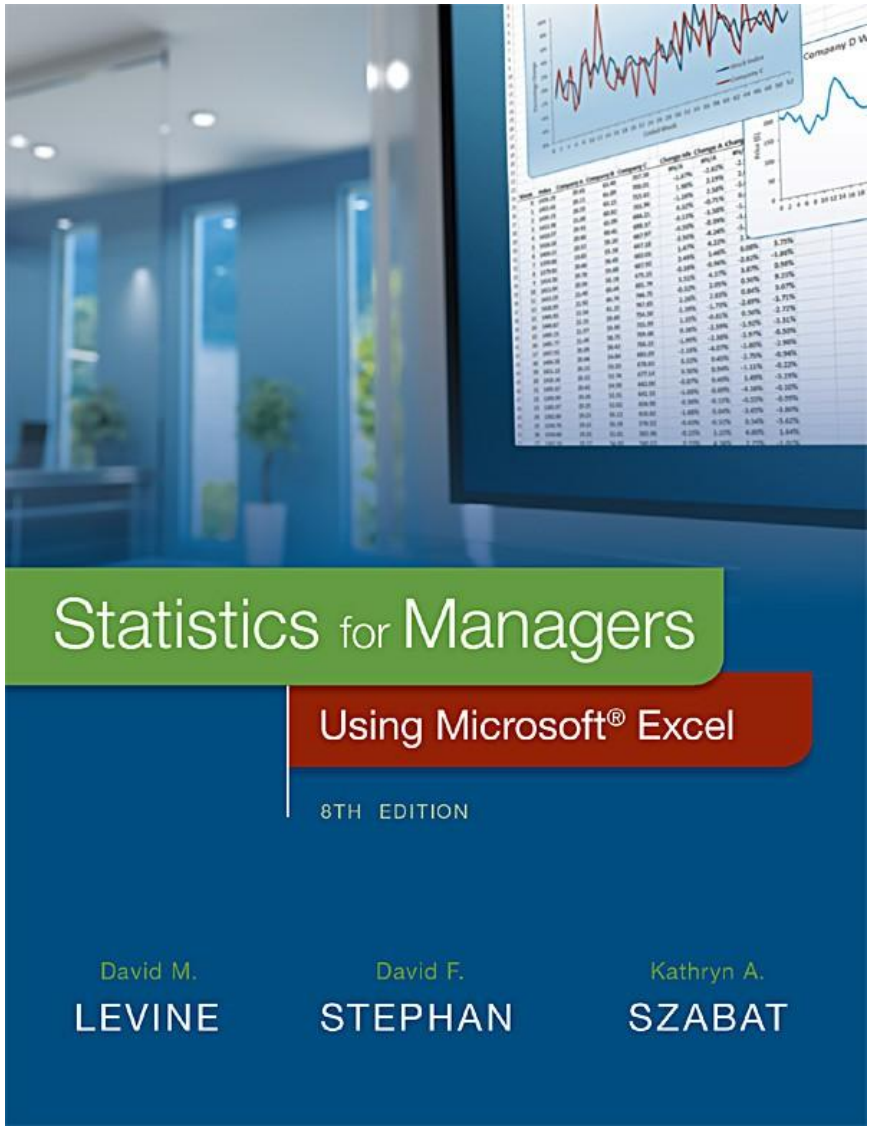


Chapter 8

Confidence Interval Estimation (Complete)

Paul Andrew Gorgonio

Objectives

In this chapter, you learn:

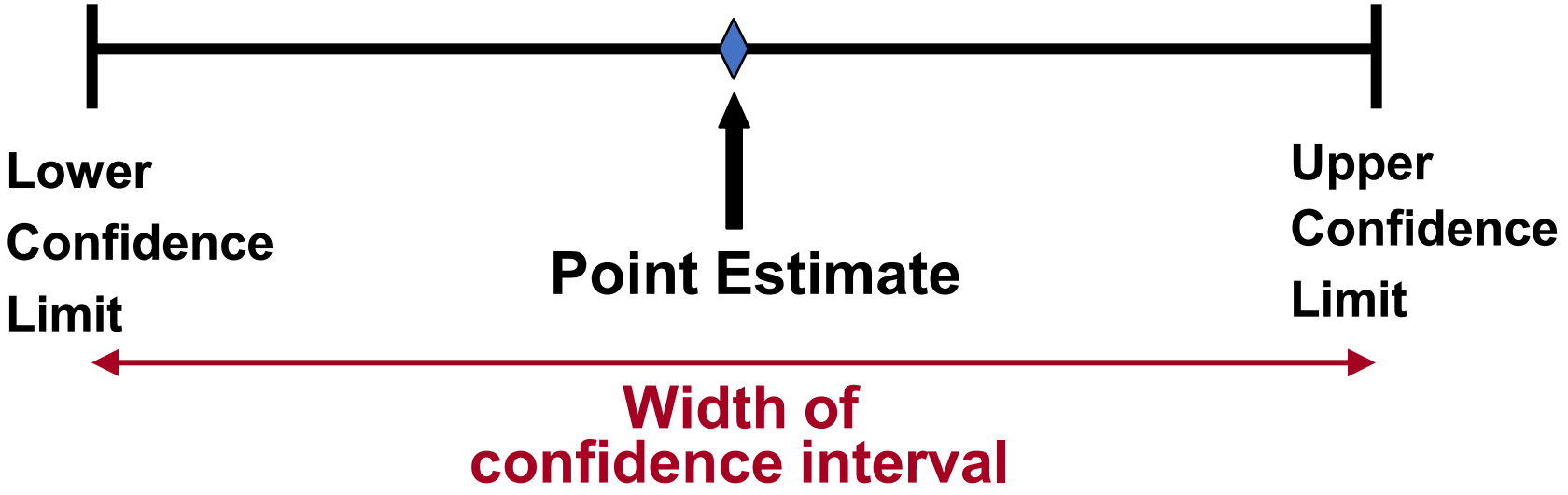
1. To construct and interpret confidence interval estimates for the population mean and the population proportion.
2. To determine the sample size necessary to develop a confidence interval for the population mean or population proportion.

Content of this Chapter

- Confidence Intervals for the Population Mean, μ:
  - when Population Standard Deviation σ is Known.
  - when Population Standard Deviation σ is Unknown.
- Confidence Intervals for the Population Proportion, π.
- Determining the Required Sample Size

Point and Interval Estimates

- A point estimate is a single number.
- A confidence interval provides additional information about the variability of the estimate.



Point Estimates

We can estimate a Population Parameter …		with a Sample Statistic (a Point Estimate)
Mean	μ	$\bar{X}$
Proportion	π	p

Confidence Intervals

- How much uncertainty is associated with a point estimate of a population parameter?
- An interval estimate provides more information about a population characteristic than does a point estimate.
- Such interval estimates are called confidence intervals.

Confidence Interval Estimate

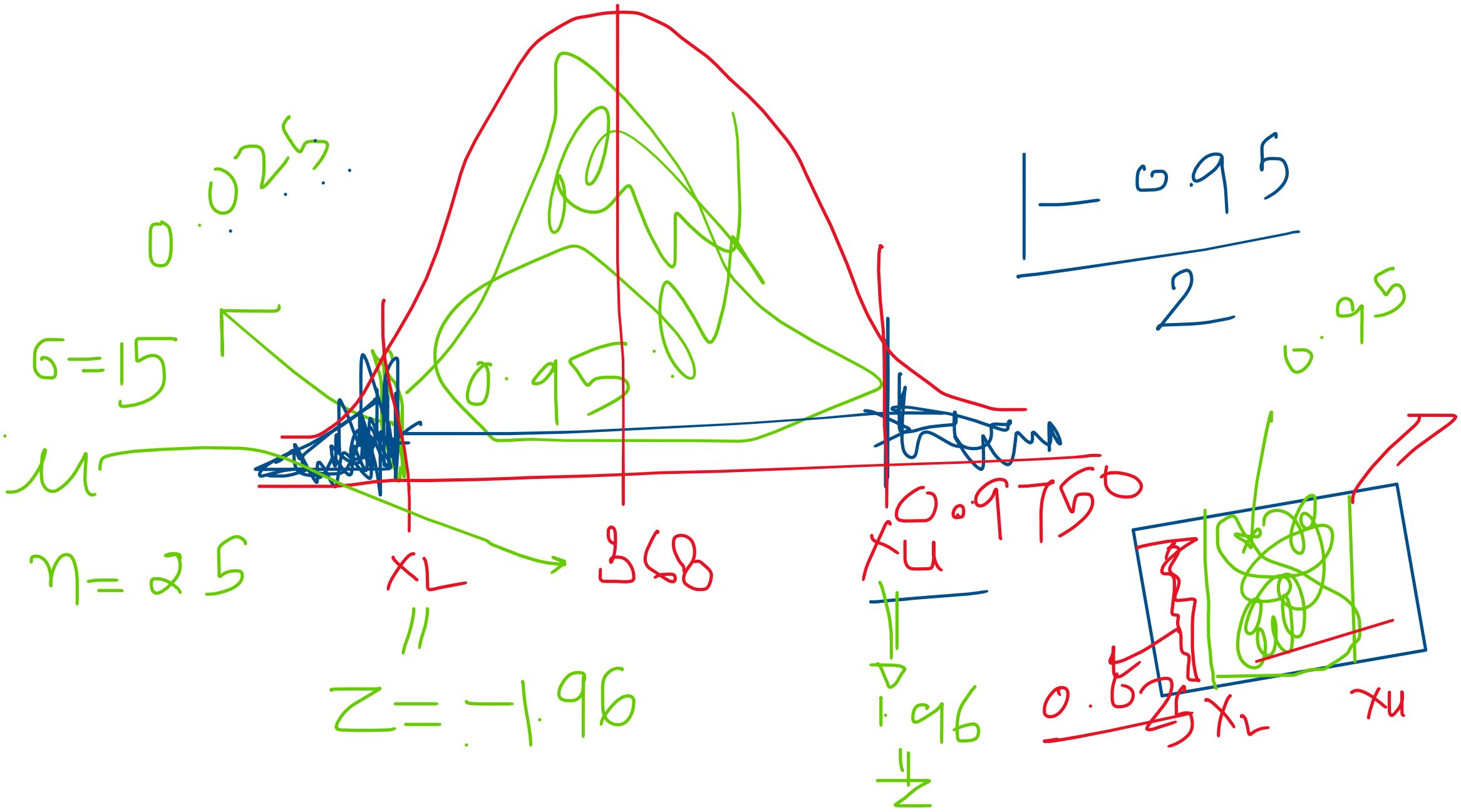
- An interval gives a range of values:
  - Takes into consideration variation in sample statistics from sample to sample.
  - Based on observations from 1 sample.
  - Gives information about closeness to unknown population parameters.
  - Stated in terms of level of confidence:
    - e.g. 95% confident, 99% confident.
    - Can never be 100% confident.

Confidence Interval Example

Cereal fill example

- Population has $\mu = 368$ and $\sigma = 15$.
- If you take a sample of size $n = 25$ you know:
  - $368 \pm 1.96 * 15 / \sqrt{25} = (362.12, 373.88)$. 95% of the intervals formed in this manner will contain μ.
  - When you don't know μ, you use $\bar{X}$ to estimate μ.
    - If $\bar{X} = 362.3$ the interval is $362.3 \pm 1.96 * 15 / 25 = (356.42, 368.18)$.
    - Since $356.42 \leq \mu \leq 368.18$ the interval based on this sample makes a correct statement about μ.

What about the intervals from other possible samples of size 25?


0.025
σ=15
μ
n=25
0.95
368
XL
XU
0.9750
1.96
Z=-1.96
1-0.95
2
0.95
0.025
XL
XU

$$X_L = Z_L \sigma_{\bar{x}} + \mu$$

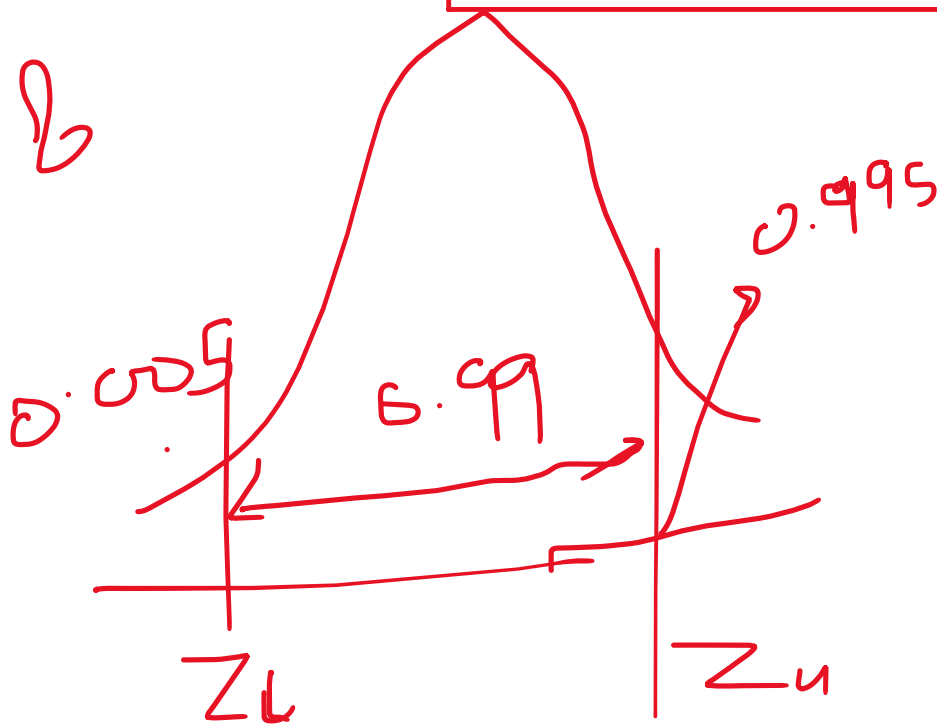
$$= -1.96\left(\frac{15}{\sqrt{25}}\right) + 368$$

$$= 362.12$$

$$X_U = 1.96 \times \frac{15}{\sqrt{25}} + 368$$

$$= 373.88$$

99



Confidence Interval Example

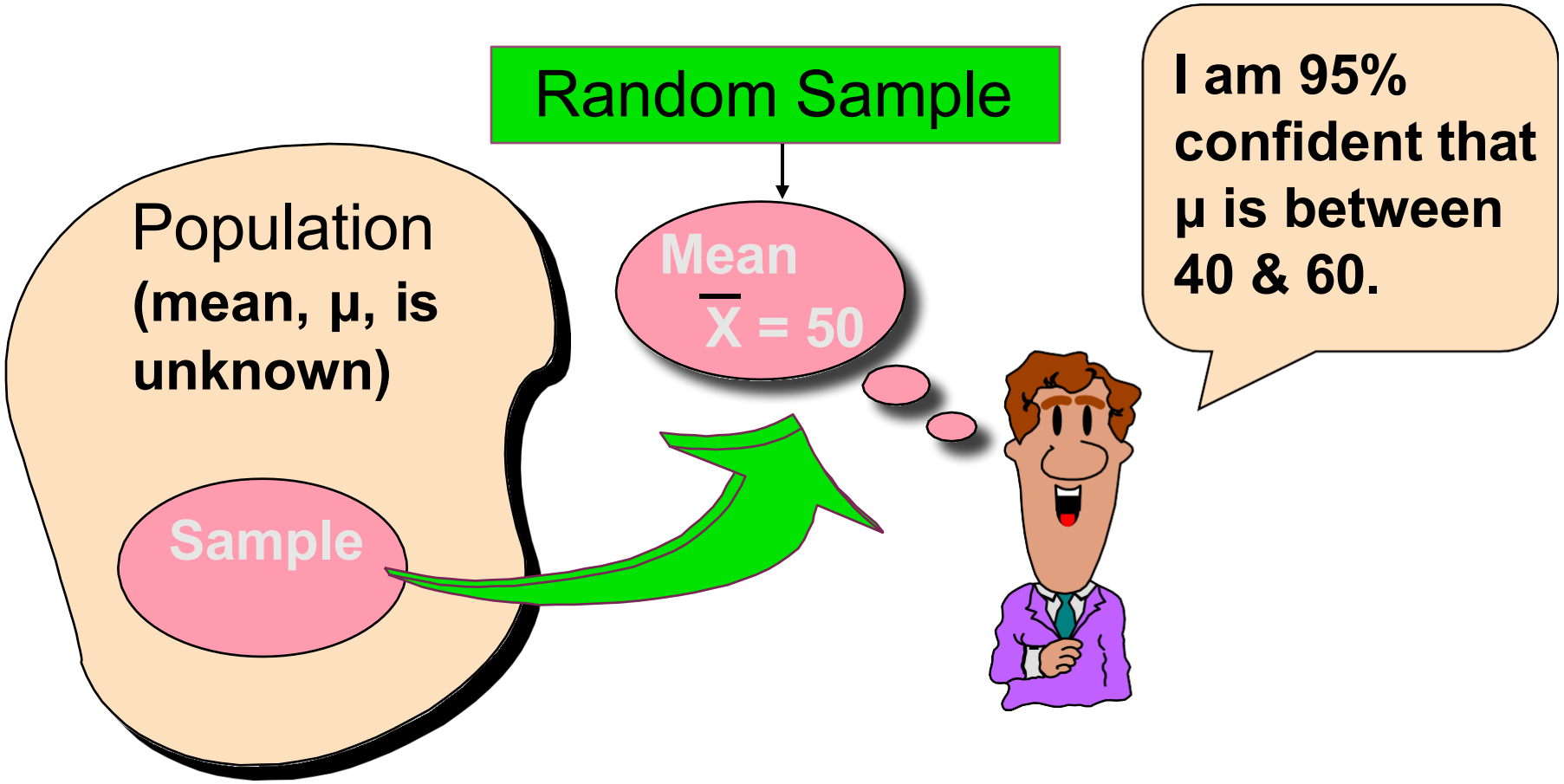
Sample #	$\overline{X}$	Lower Limit	Upper Limit	Contain μ?
1	362.30	356.42	368.18	Yes
2	369.50	363.62	375.38	Yes
3	360.00	354.12	365.88	No
4	362.12	356.24	368.00	Yes
5	373.88	368.00	379.76	Yes

Confidence Interval Example

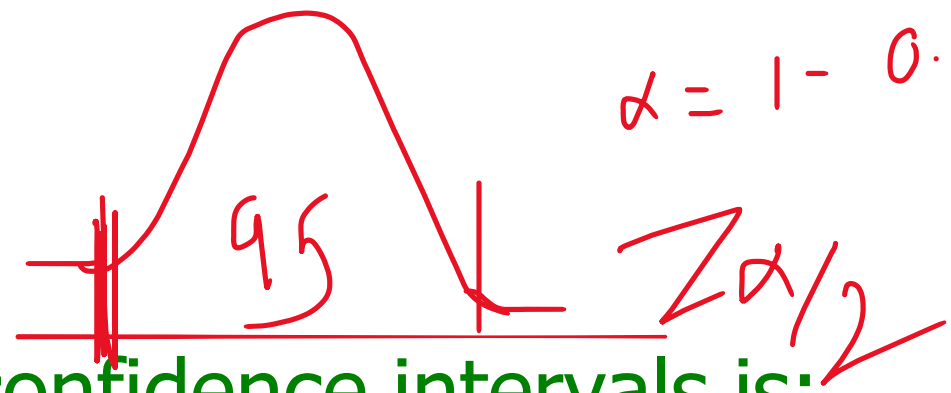
- In practice you only take one sample of size n.
- In practice you do not know μ so you do not know if the interval actually contains μ.
- However you do know that 95% of the intervals formed in this manner will contain μ.
- Thus, based on the one sample you actually selected, you can be 95% confident your interval will contain μ (this is a 95% confidence interval).

Note: 95% confidence is based on the fact that we used Z = 1.96.

Estimation Process



General Formula



- The general formula for all confidence intervals is:

Point Estimate ± (Critical Value)(Standard Error)

$\bar{X} \pm Z_{\alpha/2} \sigma/\sqrt{n}$

Where:

- Point Estimate is the sample statistic estimating the population parameter of interest.
- Critical Value is a table value based on the sampling distribution of the point estimate and the desired confidence level.
- Standard Error is the standard deviation of the point estimate.

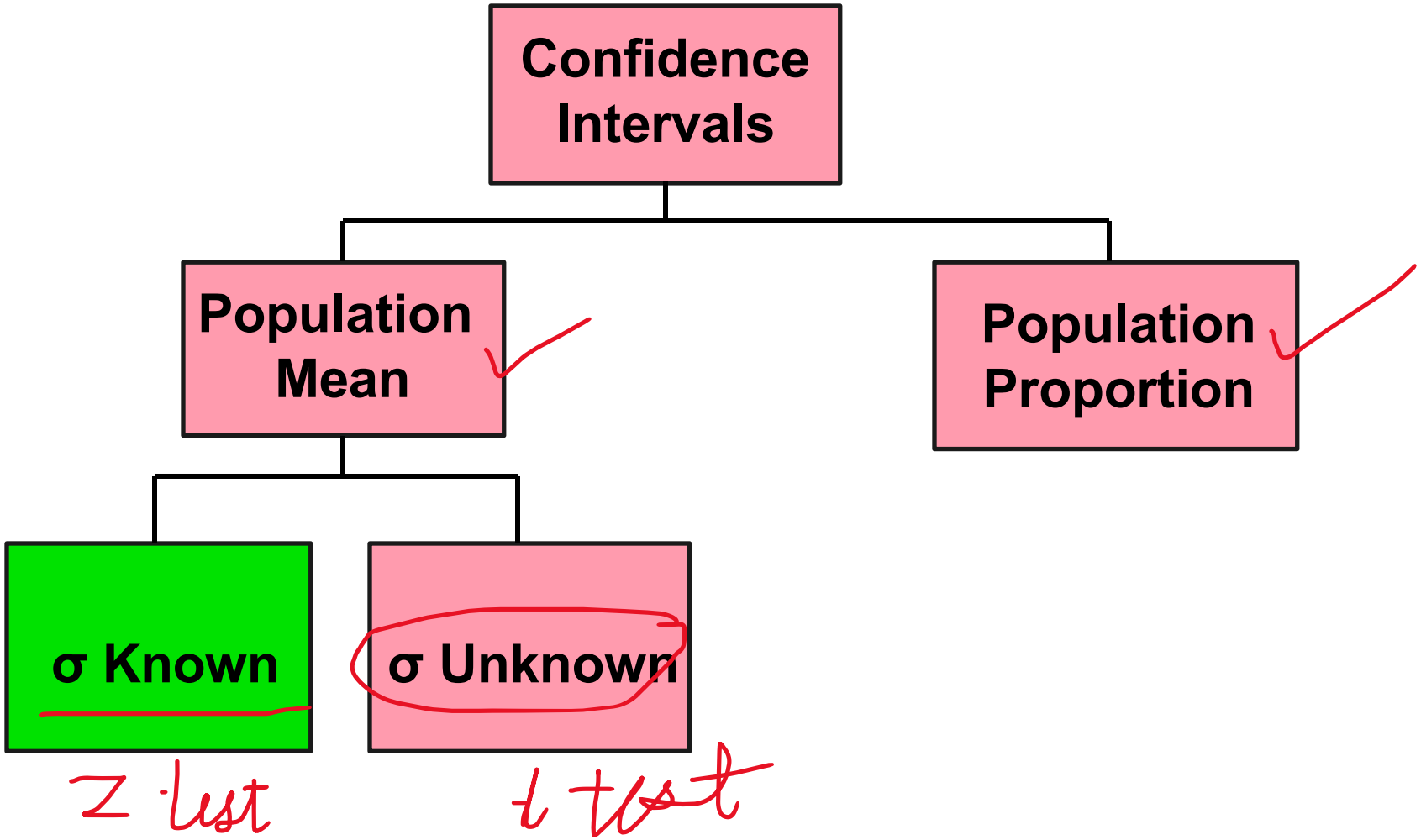
Confidence Level

- Confidence the interval will contain the unknown population parameter.
- A percentage (less than 100%).

Confidence Level, (1-α)

- Suppose confidence level = 95%.
- Also written (1 - α) = 0.95, (so α = 0.05).
- A relative frequency interpretation:
  - 95% of all the confidence intervals that can be constructed will contain the unknown true parameter.
- A specific interval either will contain or will not contain the true parameter:
  - No probability involved in a specific interval.

Confidence Intervals



Confidence Interval for μ (σ Known)

- Assumptions:
  - Population standard deviation σ is known.
  - Population is normally distributed.
  - If population is not normal, use large sample (n > 30).

- Confidence interval estimate:

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where

$\bar{X}$ is the point estimate

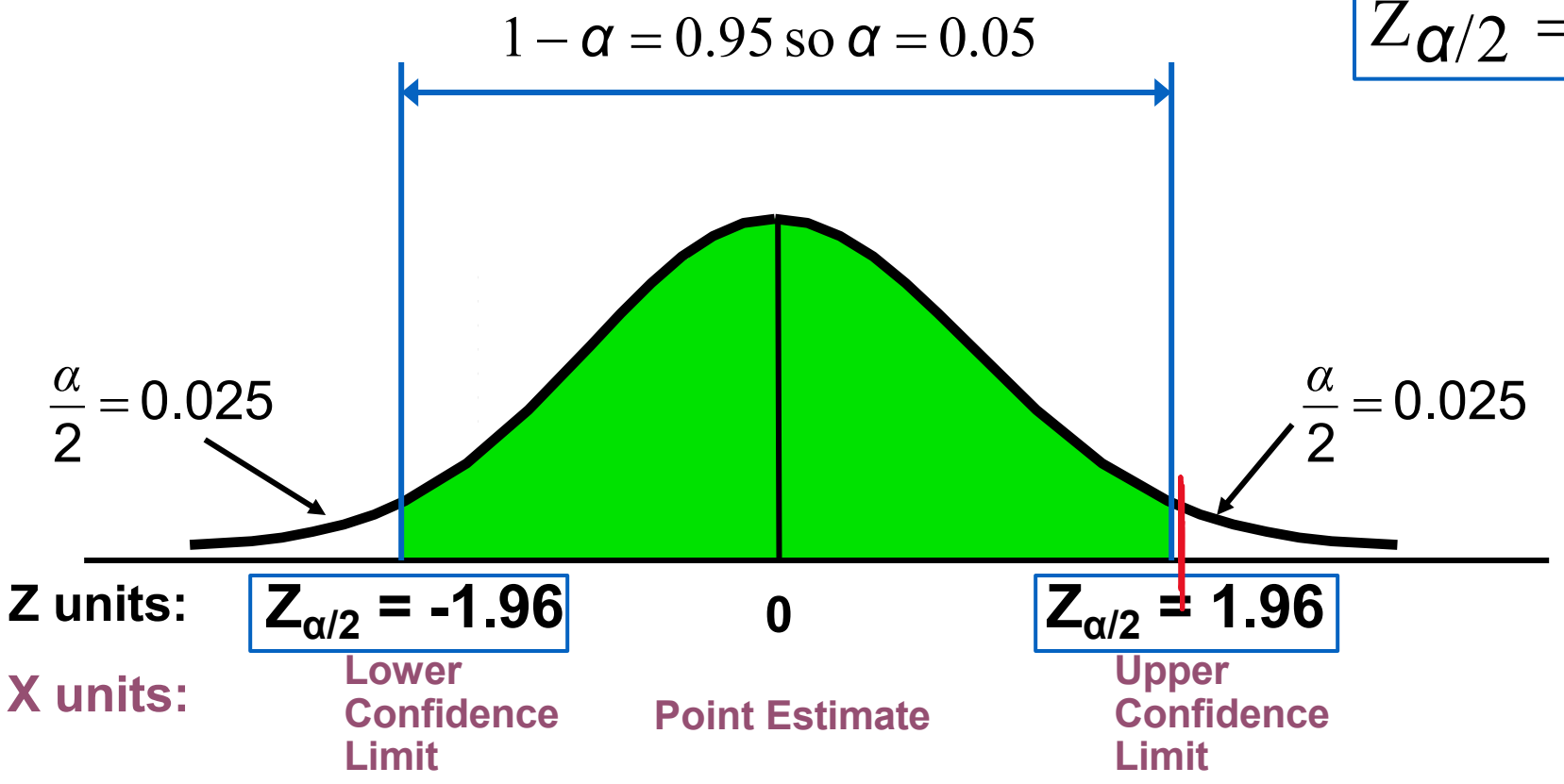
$Z_{\alpha/2}$ is the normal distribution critical value for a probability of $\alpha/2$ in each tail

$\sigma/\sqrt{n}$ is the standard error

Finding the Critical Value, $Z_{\alpha/2}$

- Consider a 95% confidence interval:

$Z_{\alpha/2} = \pm 1.96$

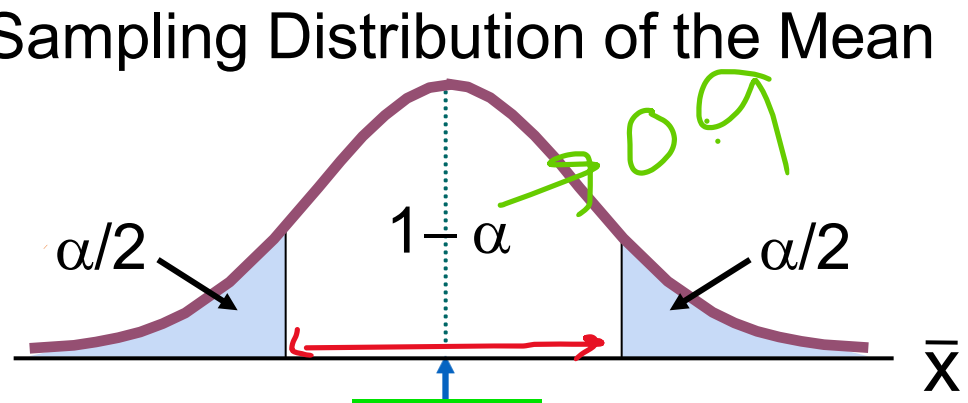


Common Levels of Confidence

- Commonly used confidence levels are 90%, 95%, and 99%.

Confidence Level	*Confidence Coefficient,* $1-\alpha$	$Z_{\alpha/2}$ *value*
80%	0.80	1.28
90%	0.90	1.645
95%	0.95	1.96
98%	0.98	2.33
99%	0.99	2.58
99.8%	0.998	3.08
99.9%	0.999	3.27

Intervals and Level of Confidence

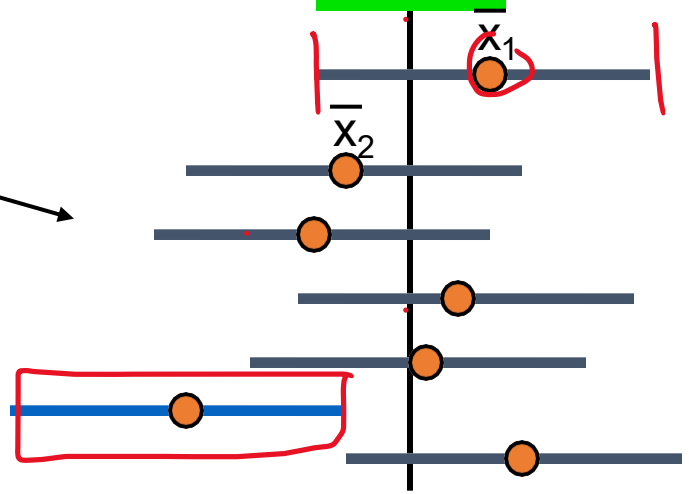


Intervals extend from

$$\overline{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

to

$$\overline{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$



Confidence Intervals

(1-α)100% of intervals constructed contain μ

(α)100% do not.

Example:

- A sample of 11 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is 0.35 ohms.

- Determine a 95% confidence interval for the true mean resistance of the population.

$m = 2.20\,\mu$

$1.99 \quad 2.40$

$n = 11$

$\sigma = 0.35$

$$X_L = 2.20 - 1.96 \times \frac{0.35}{\sqrt{11}}$$

$$X_U = 2.20 + 1.96 \times \frac{0.35}{\sqrt{11}}$$

Example:

- A sample of 11 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is 0.35 ohms.

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

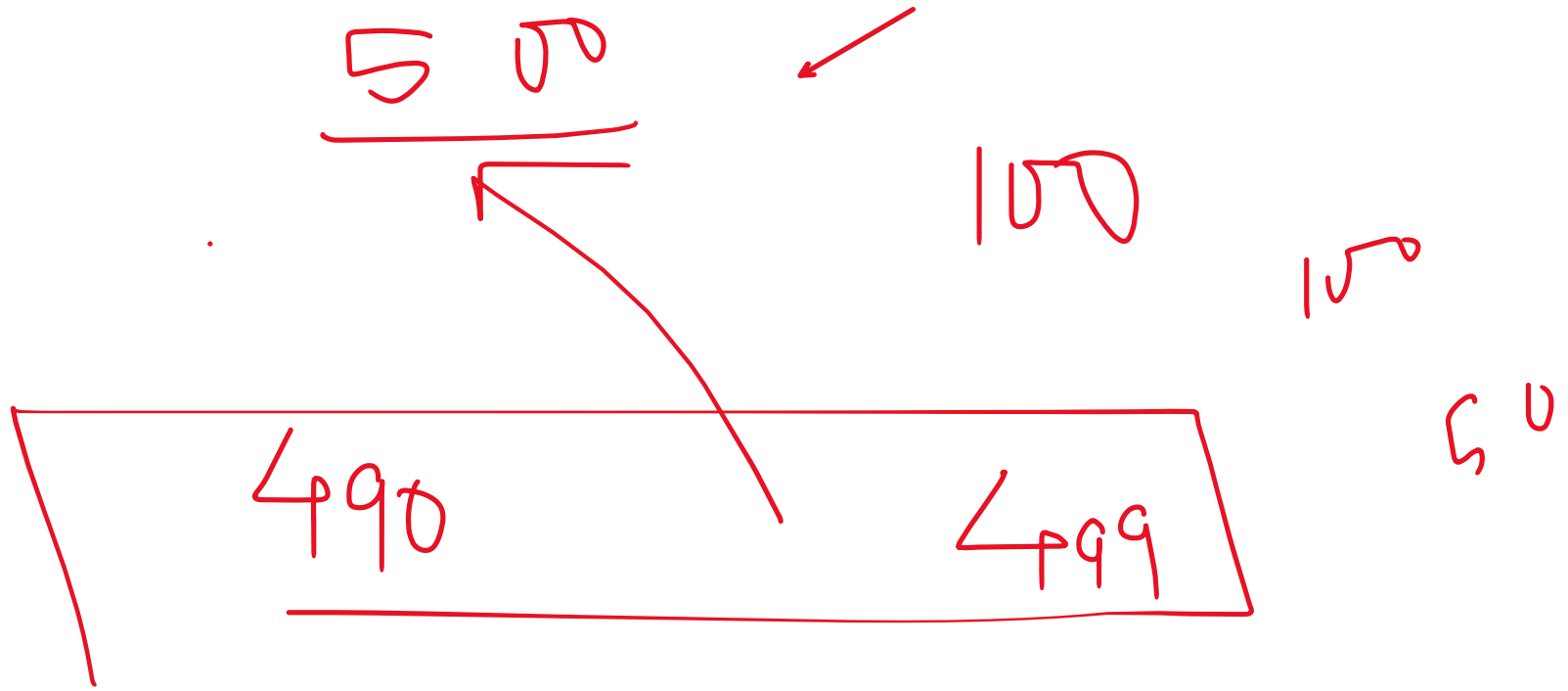
$$= 2.20 \pm 1.96\,(0.35/\sqrt{11})$$

$$= 2.20 \pm 0.2068$$

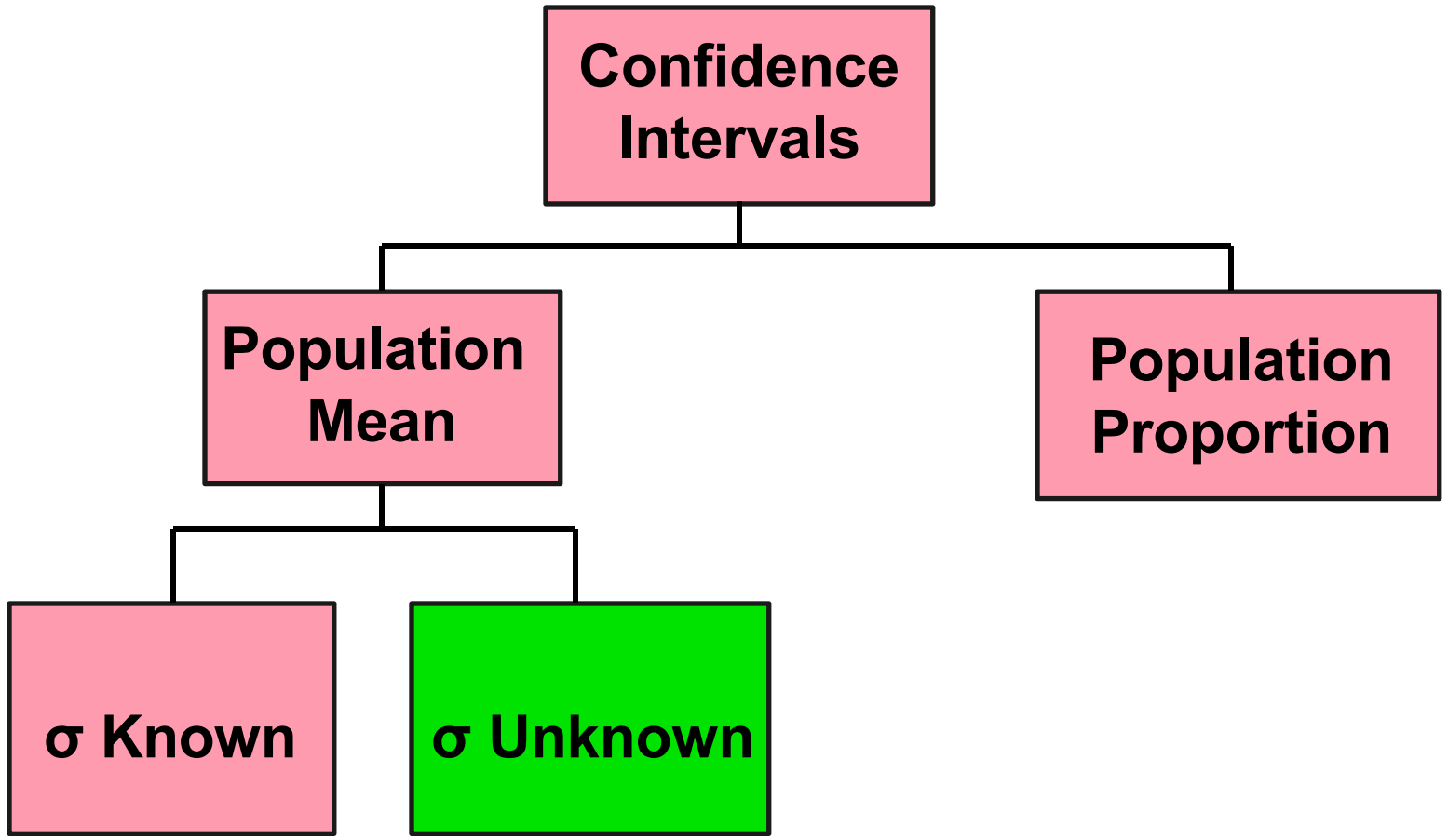
$$1.9932 \leq \mu \leq 2.4068$$

Interpretation:

- We are 95% confident that the true mean resistance is between 1.9932 and 2.4068 ohms.
- Although the true mean may or may not be in this interval, 95% of intervals formed in this manner will contain the true mean.


50
100
150
50
490
499

Confidence Intervals:



Do You Ever Truly Know σ?

- Probably not!
- In virtually all real world business situations, σ is not known.
- If there is a situation where σ is known then μ is also known (since to calculate σ you need to know μ.)
- If you truly know μ there would be no need to gather a sample to estimate it.

Confidence Interval for μ (σ Unknown)

- If the population standard deviation σ is unknown, we can substitute the sample standard deviation, S.
- This introduces extra uncertainty, since S is variable from sample to sample.
- So we use the t distribution instead of the normal distribution.

Confidence Interval for μ (σ Unknown)

- Assumptions:
  - Population standard deviation is unknown.
  - Population is normally distributed.
  - If population is not normal, use large sample (n > 30).
- Use Student's t Distribution.
- Confidence Interval Estimate:

$$\overline{X} \pm t_{\alpha/2} \frac{S}{\sqrt{n}}$$

(where $t_{\alpha/2}$ is the critical value of the t distribution with n -1 degrees of freedom and an area of α/2 in each tail.)

Student's t Distribution

- The t is a family of distributions.
- The $t_{\alpha/2}$ value depends on degrees of freedom (d.f.).
  - Number of observations that are free to vary after sample mean has been calculated.

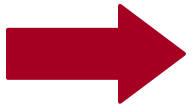
$$d.f. = n - 1$$

Degrees of Freedom (df)

Idea: Number of observations that are free to vary after sample mean has been calculated.

Example: Suppose the mean of 3 numbers is 8.0.

Let $X_1 = 7$
Let $X_2 = 8$
What is X_3?



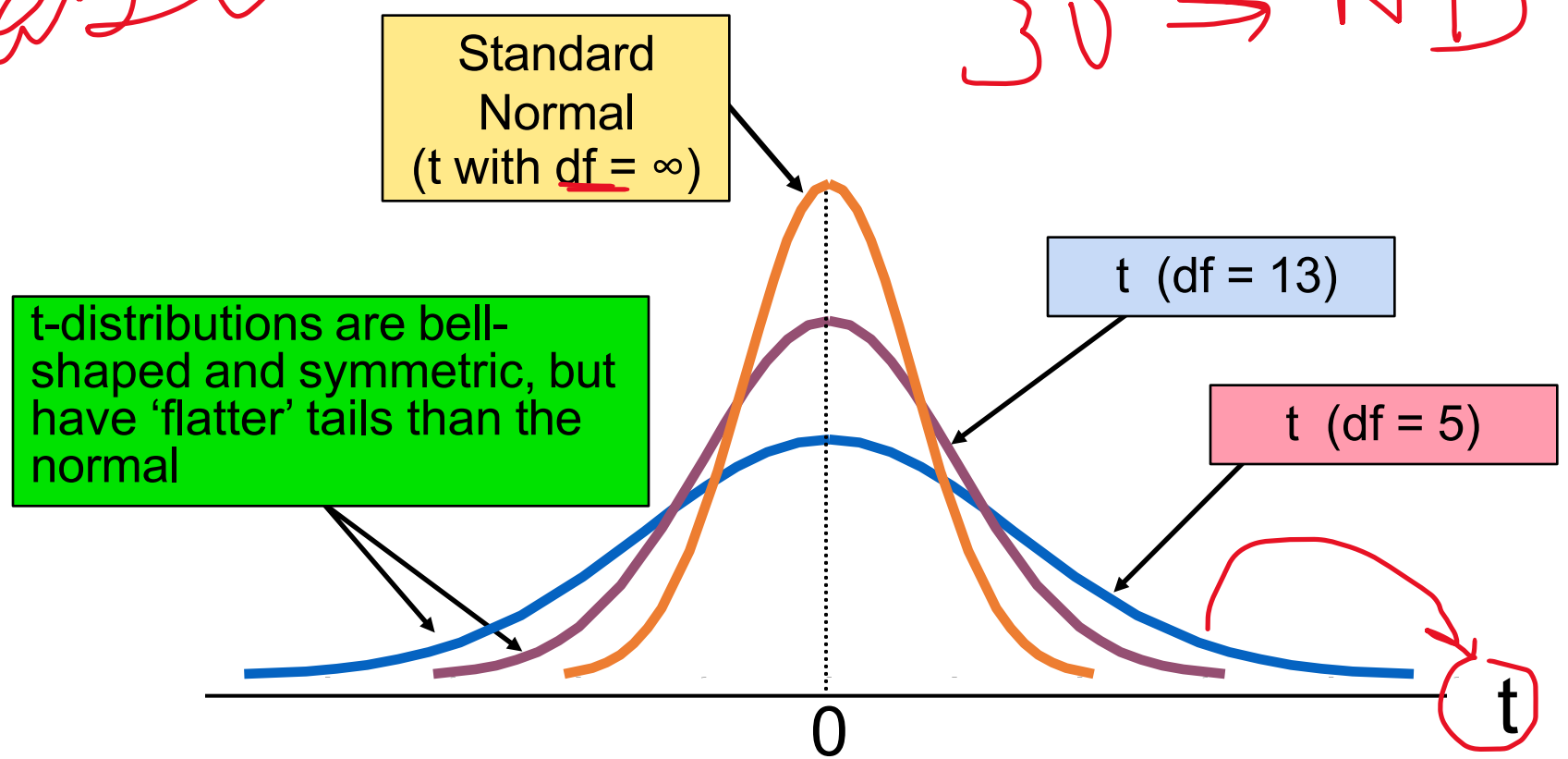
If the mean of these three values is 8.0,
then X_3 must be 9
(i.e., X_3 is not free to vary)

Here, n = 3, so degrees of freedom = n – 1 = 3 – 1 = 2.

(2 values can be any numbers, but the third is not free to vary for a given mean.)

Student's t Distribution

Note: $t \longrightarrow Z$ as n increases



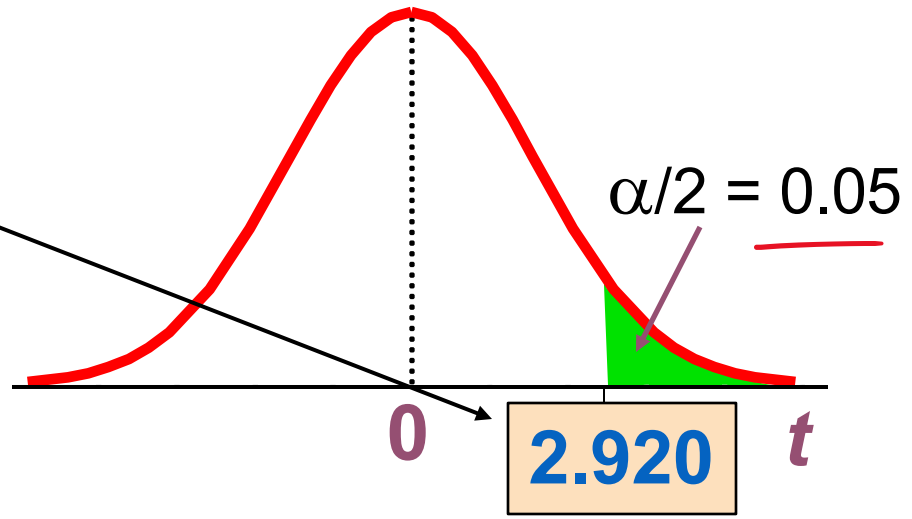
Student's t Table

	Upper Tail Area		
df	.10	**.05**	.025
1	3.078	6.314	12.706
2	1.886	**2.920**	4.303
3	1.638	2.353	3.182

The body of the table contains t values, not probabilities

Let: n = 3
df = *n* - 1 = 2
$\alpha = 0.10$
$\alpha/2 = 0.05$

CI



Selected t distribution values

With comparison to the Z value

Confidence Level	t (10 d.f.)	t (20 d.f.)	t (30 d.f.)	Z (∞ d.f.)
0.80	1.372	1.325	1.310	1.28
0.90	1.812	1.725	1.697	1.645
0.95	2.228	2.086	2.042	1.96
0.99	3.169	2.845	2.750	2.58

Note: t → Z as n increases

Example of t distribution confidence interval

A random sample of n = 25 has $\overline{X}$ = 50 and S = 8. Form a 95% confidence interval for μ.

- d.f. = n – 1 = 24, so $t_{\alpha/2} = t_{0.025} = 2.0639$

The confidence interval is

$$\overline{X} \pm t_{\alpha/2} \frac{S}{\sqrt{n}} = 50 \pm (2.0639) \frac{8}{\sqrt{25}}$$

$$46.698 \le \mu \le 53.302$$

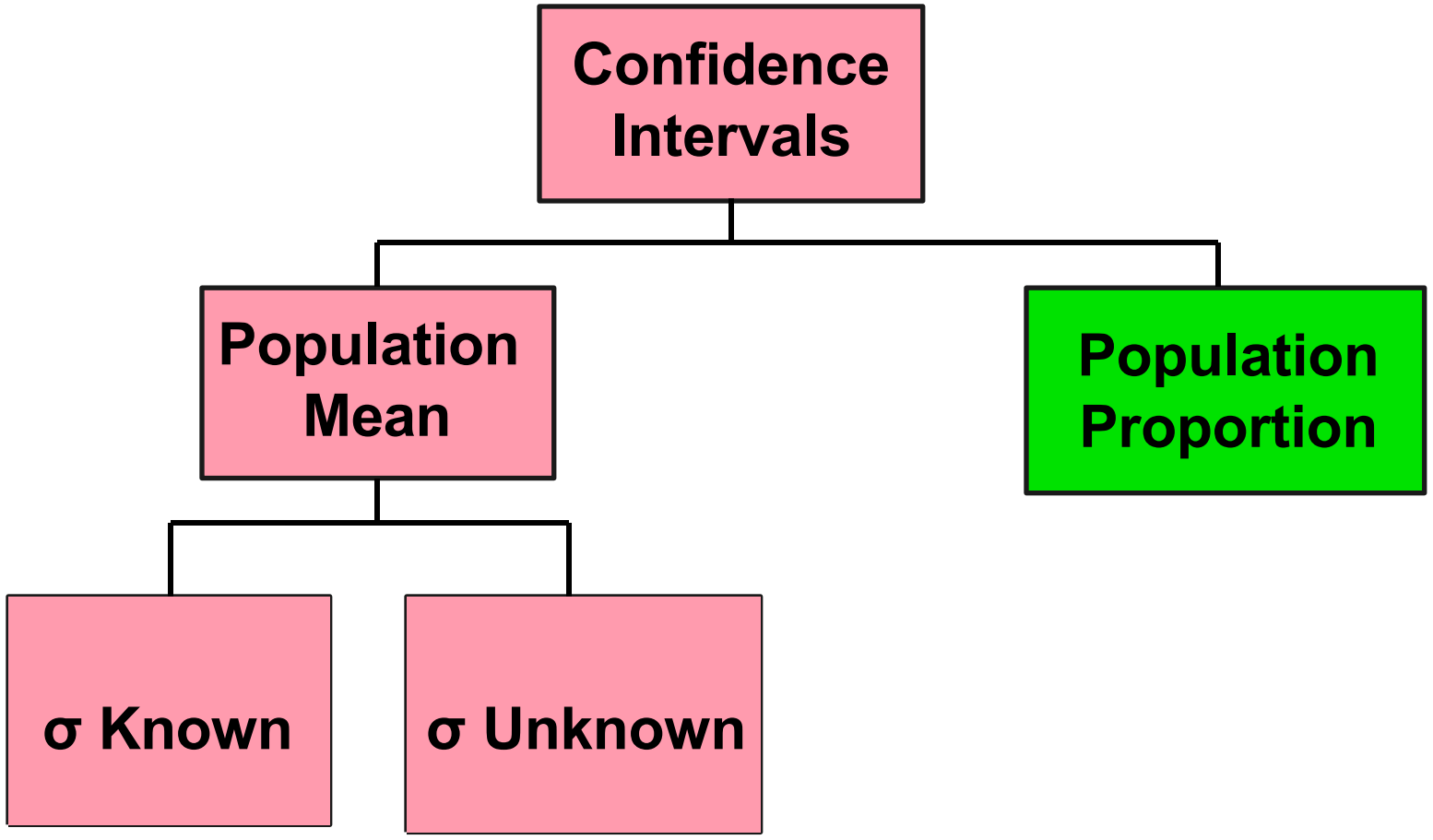
Example of t distribution confidence interval

- Interpreting this interval requires the assumption that the population you are sampling from is approximately a normal distribution (especially since n is only 25).
- This condition can be checked by creating a:
  - Normal probability plot or
  - Boxplot.

Using Excel for a t-interval for μ

	A	B	
1	Confidence Interval Estimate for the Mean		
2			
3	Data		
4	Sample Standard Deviation	28.95	
5	Sample Mean	110.27	
6	Sample Size	100	
7	Confidence Level	95%	
8			
9	Intermediate Calculations		
10	Standard Error of the Mean	2.895	=B4/SQRT(B6)
11	Degrees of Freedom	99	=B6 - 1
12	*t* Value	1.9842	=T.INV.2T(1 - B7, B11)
13	Interval Half Width	5.7443	=B12 * B10
14			
15	Confidence Interval		
16	Interval Lower Limit	104.53	=B5 - B13
17	Interval Upper Limit	116.01	=B5 + B13

Confidence Intervals



Confidence Intervals for the Population Proportion, π

- An interval estimate for the population proportion (π) can be calculated by adding an allowance for uncertainty to the sample proportion (p).

Confidence Intervals for the Population Proportion, π

- Recall that the distribution of the sample proportion is approximately normal if the sample size is large, with standard deviation:

$$\sigma_p = \sqrt{\frac{\pi(1-\pi)}{n}}$$

- We will estimate this with sample data:

$$\sqrt{\frac{p(1-p)}{n}}$$

Confidence Interval Endpoints

- Upper and lower confidence limits for the population proportion are calculated with the formula:

$$p \pm Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

- where
  - $Z_{\alpha/2}$ is the standard normal value for the level of confidence desired
  - p is the sample proportion
  - n is the sample size.
- Note: must have $np > 5$ and $n(1-p) > 5$.

Example:

- A random sample of 100 people shows that 25 are left-handed.
- Form a 95% confidence interval for the true proportion of left-handers.

- Solution:

$$p \pm Z_{\alpha/2}\sqrt{p(1-p)/n}$$

$$= 25/100 \pm 1.96\sqrt{0.25(0.75)/100}$$

$$= 0.25 \pm 1.96(0.0433)$$

$$= 0.1651 \leq \pi \leq 0.3349$$

Interpretation:

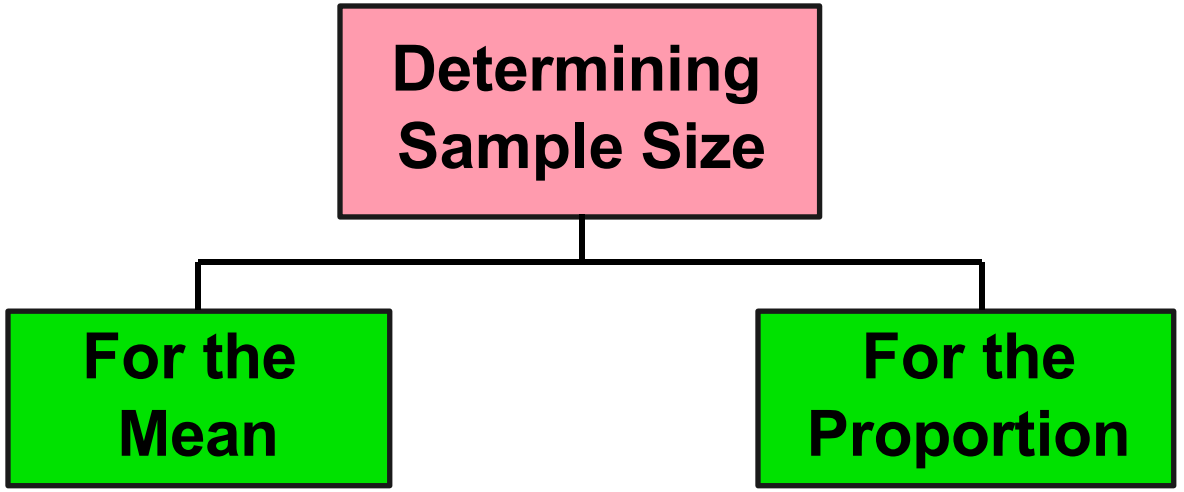
- We are 95% confident that the true percentage of left-handers in the population is between 16.51% and 33.49%.

- Although the interval from 0.1651 to 0.3349 may or may not contain the true proportion, 95% of intervals formed from samples of size 100 in this manner will contain the true proportion.

Using Excel for a Confidence Interval for π

	A	B	
1	**Proportion of In-Error Sales Invoices**		
2			
3	**Data**		
4	Sample Size	100	
5	Number of Successes	10	
6	Confidence Level	95%	
7			
8	Intermediate Calculations		
9	Sample Proportion	0.1	=B5/B4
10	Z Value	-1.9600	=NORM.S.INV((1 - B6)/2)
11	Standard Error of the Proportion	0.03	=SQRT(B9 * (1 - B9)/B4)
12	Interval Half Width	0.0588	=ABS(B10 * B11)
13			
14	**Confidence Interval**		
15	Interval Lower Limit	0.0412	=B9 - B12
16	Interval Upper Limit	0.1588	=B9 + B12

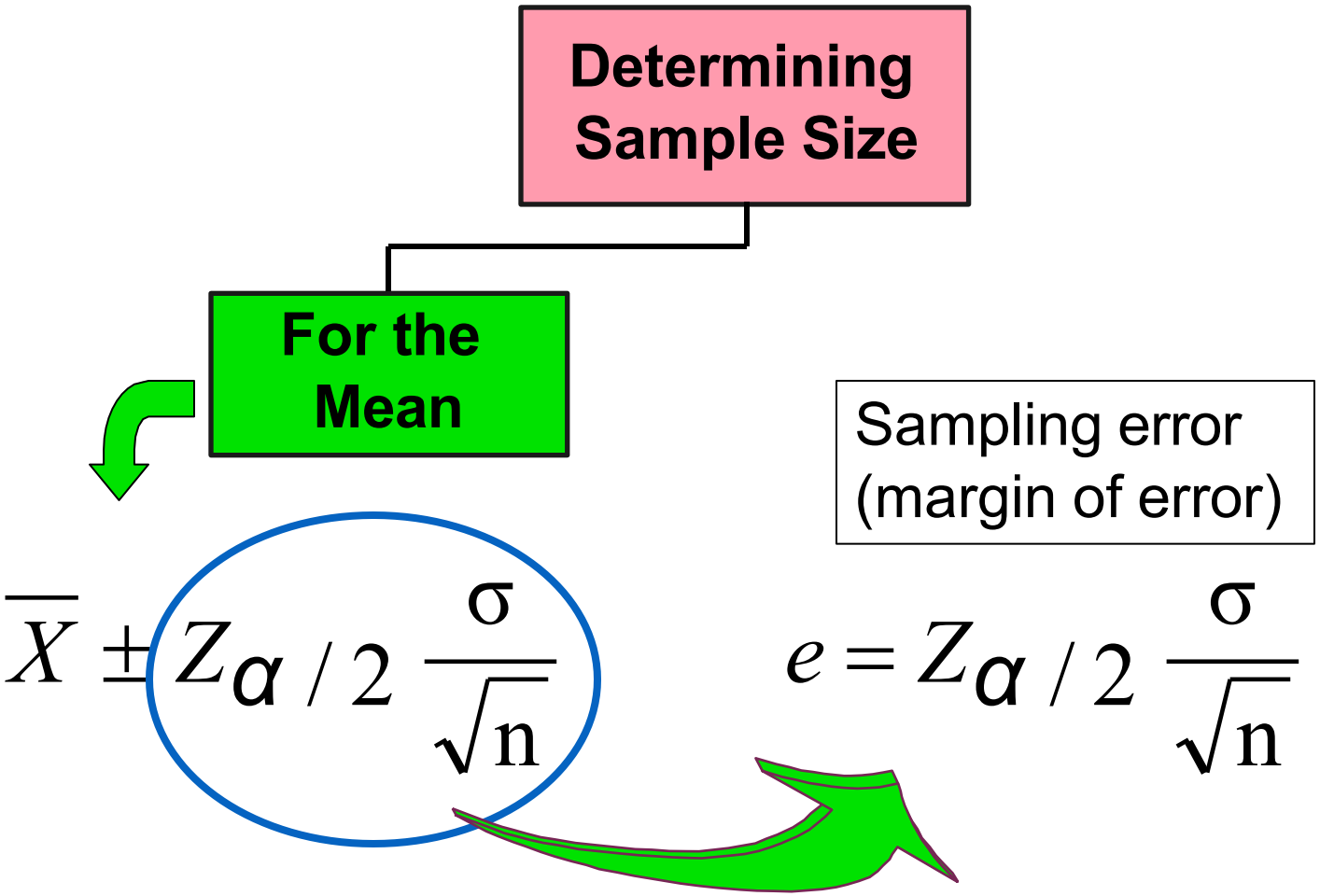
Determining Sample Size



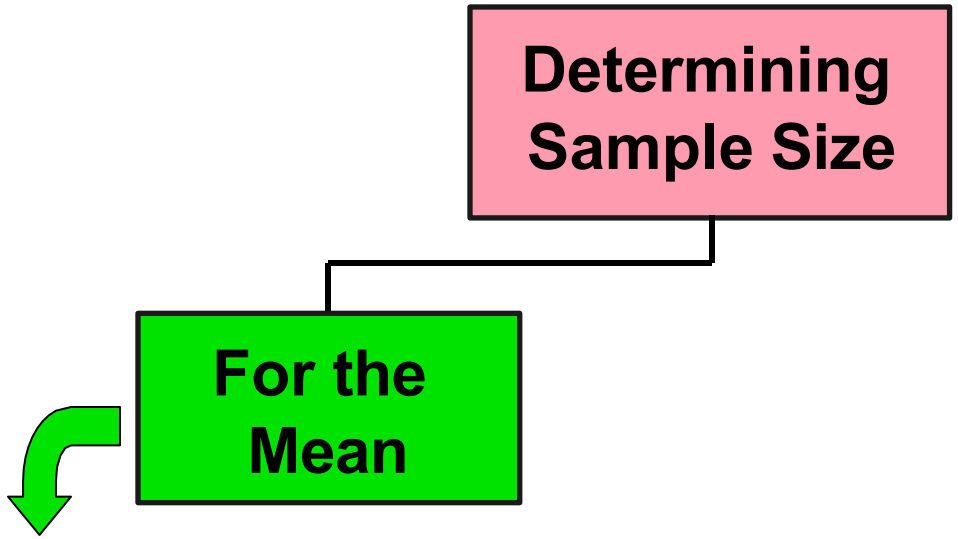
Sampling Error

- The required sample size can be found to reach a desired margin of error (e) with a specified level of confidence (1 - α).
- The margin of error is also called sampling error:
  - the amount of imprecision in the estimate of the population parameter.
  - the amount added and subtracted to the point estimate to form the confidence interval.

Determining Sample Size



Determining Sample Size



$$e = Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$ → Now solve for n to get →

$$n = \frac{Z_{\alpha/2}^2 \sigma^2}{e^2}$$

Determining Sample Size

- To determine the required sample size for the mean, you must know:
  - The desired level of confidence $(1 - \alpha)$, which determines the critical value, $Z_{\alpha/2}$.
  - The acceptable sampling error, e.
  - The standard deviation, σ.

Required Sample Size Example

If $\sigma = 45$, what sample size is needed to estimate the mean within ± 5 with 90% confidence?

$$n = \frac{Z^2\ \sigma^2}{e^2} = \frac{(1.645)^2(45)^2}{5^2} = 219.19$$

So the required sample size is **n = 220**

(Always round up)

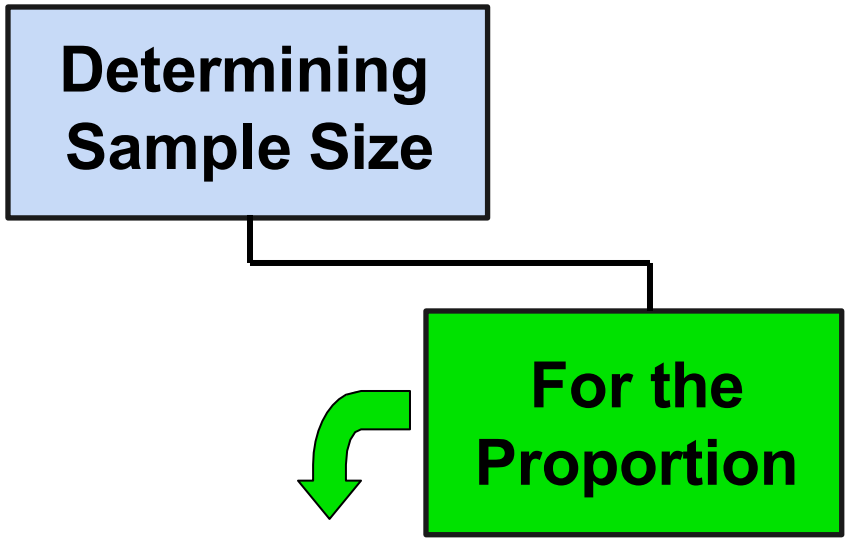
Using Excel To Calculate Sample Size For The Mean

	A	B	
1	For the Mean Sales Invoice Amount		
2			
3	Data		
4	Population Standard Deviation	25	
5	Sampling Error	5	
6	Confidence Level	95%	
7			
8	Intemediate Calculations		
9	Z Value	-1.9600	=NORM.S.INV((1 - B6)/2)
10	Calculated Sample Size	96.0365	=((B9 * B4)/B5)^2
11			
12	Result		
13	Sample Size Needed	97	=ROUNDUP(B10, 0)

If σ is unknown

- If unknown, σ can be estimated when using the required sample size formula.
  - Use a value for σ that is expected to be at least as large as the true σ.
  - Select a pilot sample and estimate σ with the sample standard deviation, S.

Determining Sample Size



$$e = Z\sqrt{\frac{\pi(1-\pi)}{n}} \rightarrow \text{Now solve for n to get} \rightarrow n = \frac{Z^2_{\alpha/2}\,\pi(1-\pi)}{e^2}$$

Determining Sample Size

- To determine the required sample size for the proportion, you must know:
  - The desired level of confidence (1 - α), which determines the critical value, $Z_{\alpha./2}$.
  - The acceptable sampling error, e.
  - The true proportion of events of interest, π.
    - π can be estimated with a pilot sample if necessary (or conservatively use 0.5 as an estimate of π.)

Required Sample Size Example

How large a sample would be necessary to estimate the true proportion of sales invoices containing errors in a large population within ±7%, with 95% confidence?

(Assume a pilot sample yields $p = 0.15$.)

Required Sample Size Example

Solution:

For 95% confidence, use $Z_{\alpha/2} = 1.96$

e = 0.07

p = 0.15, so use this to estimate π.

$$n = \frac{Z^2_{\alpha/2}\pi(1-\pi)}{e^2} = \frac{(1.96)^2(0.15)(1-0.15)}{(0.07)^2} = 99.96$$

So use n = 100

Using Excel To Calculate A Sample Size For A Proportion

How large a sample would be necessary to estimate the true proportion of sales invoices containing errors in a large population within ±7%, with 95% confidence?

(Assume a pilot sample yields p = 0.15.)

	A	B	
1	For the Proportion of In-Error Sales Invoices		
2			
3	Data		
4	Estimate of True Proportion	0.15	
5	Sampling Error	0.07	
6	Confidence Level	95%	
7			
8	Intermediate Calculations		
9	Z Value	-1.9600	=NORM.S.INV((1 - B6) / 2)
10	Calculated Sample Size	99.9563	=(B9^2 * B4 * (1 - B4)) / B5^2
11			
12	Result		
13	Sample Size Needed	100	=ROUNDUP(B10, 0)

Ethical Issues

- A confidence interval estimate (reflecting sampling error) should always be included when reporting a point estimate.
- The level of confidence should always be reported.
- The sample size should be reported.
- An interpretation of the confidence interval estimate should also be provided.

Chapter Summary

In this chapter we discussed:

- The construction and interpretation of confidence interval estimates for the population mean and the population proportion.
- The determination of the sample size necessary to develop a confidence interval for the population mean or population proportion